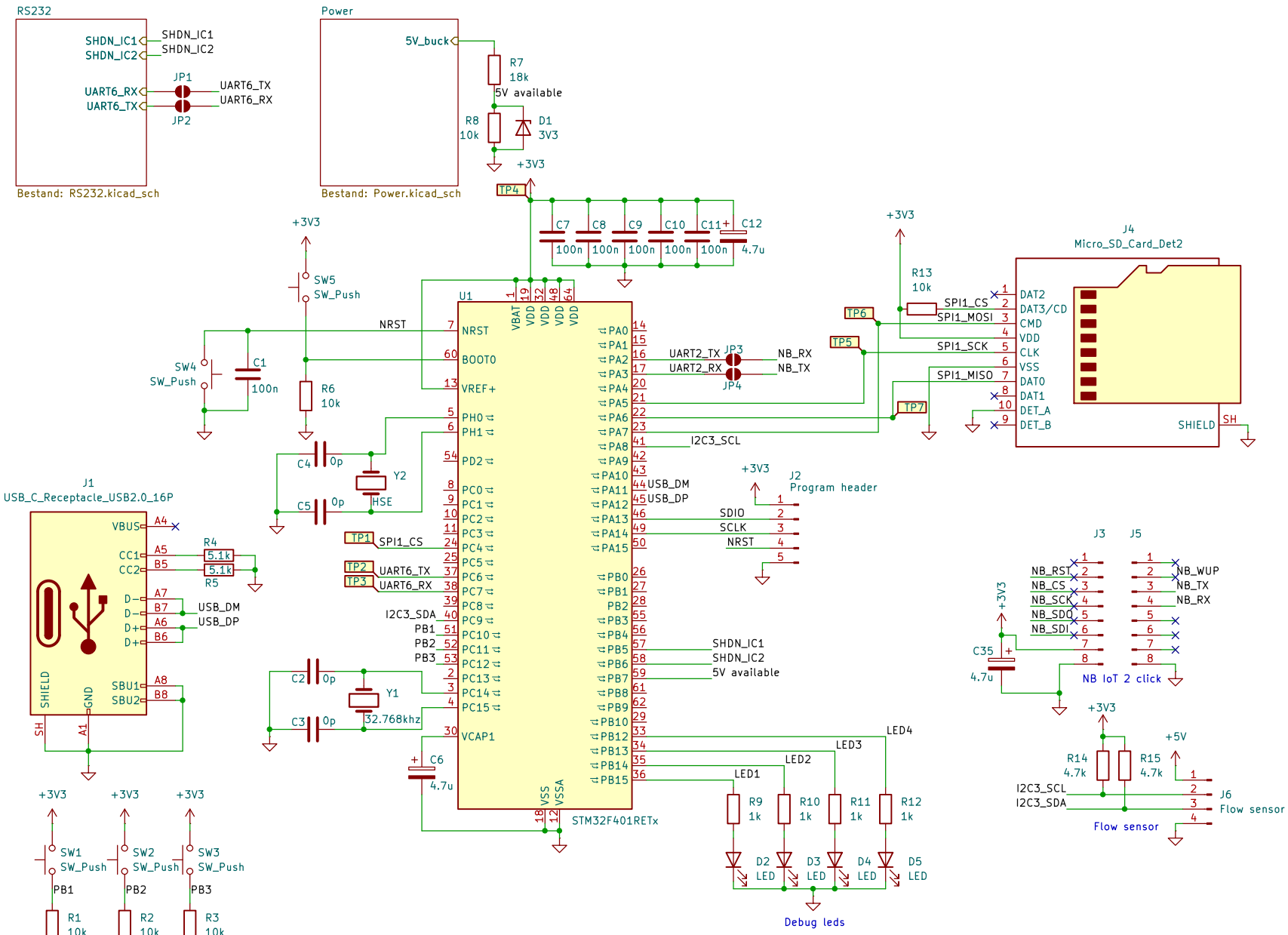
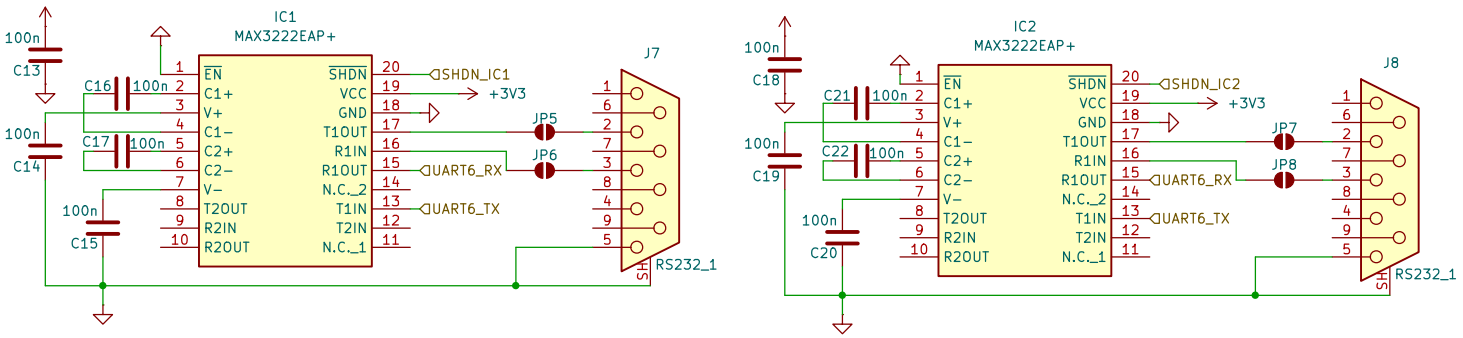
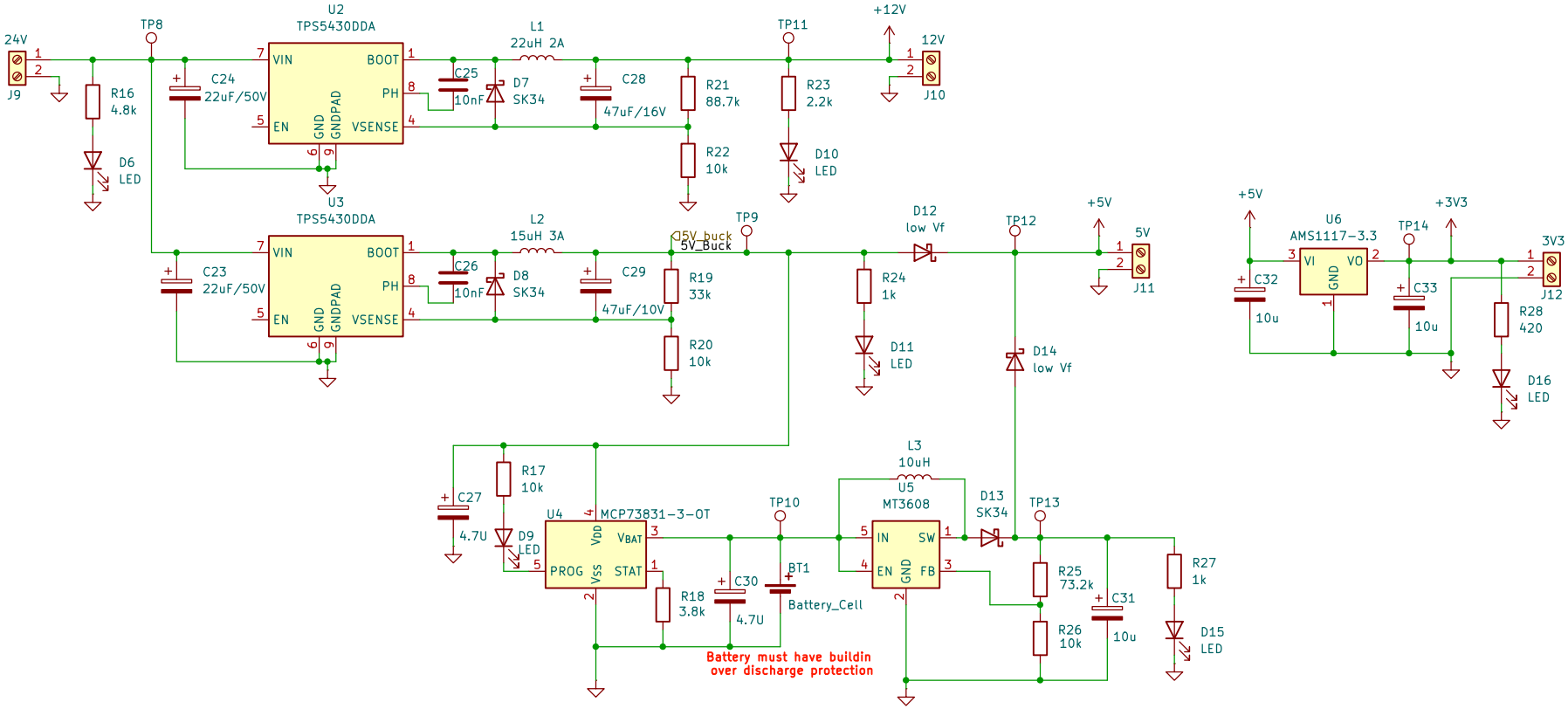

RS232
SHDN_IC1
SHDN_IC2
UART6_RX
UART6_TX
JP1
JP2
UART6_TX
UART6_RX
Bestand: RS232.kicad_sch
Power
5V_buck
R7
18k
5V available
R8
10k
D1
3V3
+3V3
Bestand: Power.kicad_sch
TP4
C7
C8
C9
C10
C11
C12
100n
100n
100n
100n
100n
4.7u
+3V3
SW5
SW_Push
NRST
SW4
SW_Push
C1
100n
R6
10k
C4
0p
C5
0p
Y2
HSE
U1
VBAT
VDD
VDD
VDD
VDD
NRST
BOOT0
VREF+
PH0
PH1
PD2
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
VCAP1
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
VSS
VSSA
STM32F401RETx
UART2_TX
UART2_RX
JP3
JP4
NB_RX
NB_TX
I2C3_SCL
USB_DM
USB_DP
SDIO
SCLK
NRST
SHDN_IC1
SHDN_IC2
5V available
LED4
LED3
LED2
LED1
J2
Program header
TP1
SPI1_CS
TP2
UART6_TX
TP3
UART6_RX
I2C3_SDA
PB1
PB2
PB3
C2
0p
C3
0p
Y1
32.768khz
C6
4.7u
J1
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
SH
A1
R4
5.1k
R5
5.1k
USB_DM
USB_DP
+3V3
SW1
SW_Push
SW2
SW_Push
SW3
SW_Push
PB1
PB2
PB3
R1
10k
R2
10k
R3
10k
R9
1k
R10
1k
R11
1k
R12
1k
D2
LED
D3
LED
D4
LED
D5
LED
Debug leds
+3V3
R13
10k
TP6
TP5
TP7
SPI1_CS
SPI1_MOSI
SPI1_SCK
SPI1_MISO
J4
Micro_SD_Card_Det2
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
SH
J3
J5
NB_RST
NB_CS
NB_SCK
NB_SDO
NB_SDI
NB_WUP
NB_TX
NB_RX
+3V3
C35
4.7u
NB IoT 2 click
+3V3
+5V
R14
4.7k
R15
4.7k
I2C3_SCL
I2C3_SDA
J6
Flow sensor
Flow sensor


IC1
MAX3222EAP+
100n
C13
C16
100n
C17
100n
100n
C14
100n
C15
EN
C1+
V+
C1−
C2+
C2−
V−
T2OUT
R2IN
R2OUT
SHDN
VCC
GND
T1OUT
R1IN
R1OUT
N.C._2
T1IN
T2IN
N.C._1
SHDN_IC1
+3V3
UART6_RX
UART6_TX
JP5
JP6
J7
RS232_1
SH
IC2
MAX3222EAP+
100n
C18
C21
100n
C22
100n
100n
C19
100n
C20
J8
JP7
JP8


24V
J9
R16
4.8k
D6
LED
TP8
U2
TPS5430DDA
VIN
BOOT
PH
EN
GND
GNDPAD
VSENSE
C24
22uF/50V
C25
10nF
L1
22uH 2A
D7
SK34
C28
47uF/16V
R21
88.7k
R22
10k
TP11
R23
2.2k
D10
LED
+12V
12V
J10
U3
TPS5430DDA
C23
22uF/50V
C26
10nF
L2
15uH 3A
D8
SK34
C29
47uF/10V
5V_buck
5V_Buck
TP9
R19
33k
R20
10k
D12
low Vf
TP12
R24
1k
D11
LED
D14
low Vf
+5V
5V
J11
+5V
U6
AMS1117-3.3
VI
VO
GND
C32
10u
TP14
+3V3
C33
10u
3V3
J12
R28
420
D16
LED
L3
10uH
U5
MT3608
IN
SW
EN
GND
FB
D13
SK34
TP13
R25
73.2k
R26
10k
C31
10u
R27
1k
D15
LED
R17
10k
C27
4.7U
D9
LED
U4
MCP73831-3-OT
VDD
VBAT
PROG
VSS
STAT
TP10
R18
3.8k
C30
4.7U
BT1
Battery_Cell
Battery must have buildin
over discharge protection